



Visual Analysis for Urban Traffic Network with Multiple CCTV Videos

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Motivation

- Traffic network modeling for urban traffic analysis is an important task to minimize traffic congestion, and VD and GPS data are frequently used.
- VD system requires expensive equipment, only detects bidirectional vehicle information and GPS trajectory data has statistical representativeness problem.

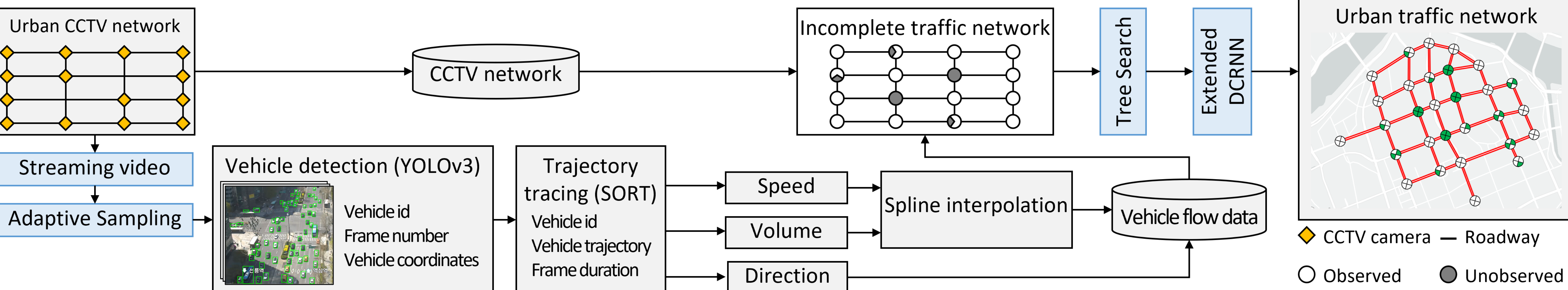


Figure 1. Our Framework

Framework

- The framework samples CCTV video data from massive streaming video and extracts vehicle flow data using YOLOv3 and SORT.
- Virtual network is automatically created by tree search algorithm. Then, the unobserved nodes are predicted using the extended DCRNN model with the virtual network as input.

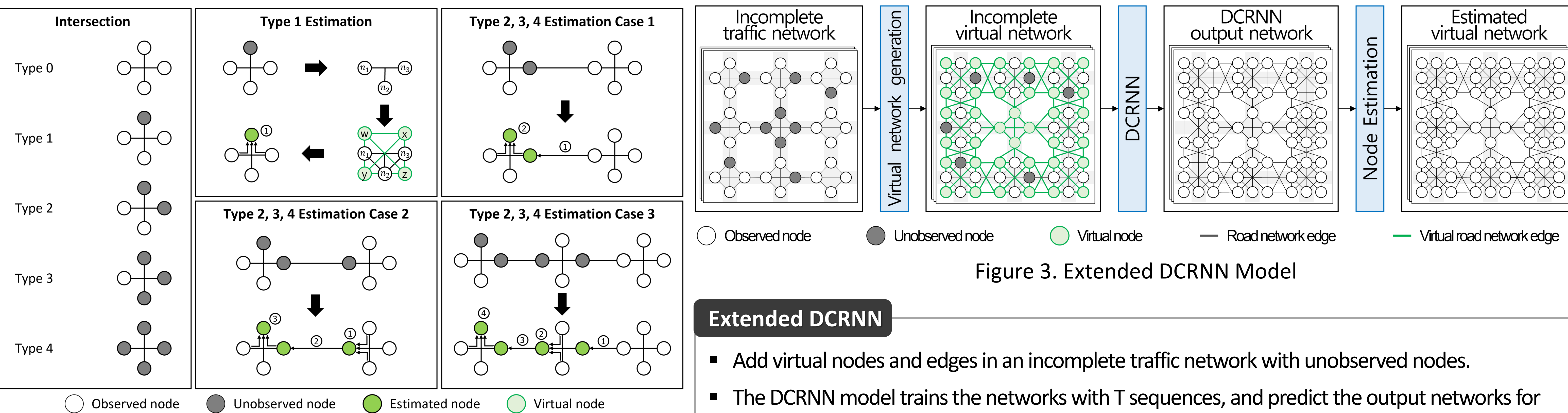


Figure 2. Tree Search Algorithm

Tree search

- We apply a tree search algorithm to pre-estimate unobserved nodes in an incomplete network for an automated virtual network generation.
- Type 1 estimates the value of the unobserved node through the values of the other three nodes.
- Types 2, 3, and 4 estimate the value of the unobserved node at the time $T + t$ through the value at the time T of the opposite node.

Time	metric	Case 1				Case 2				
		Unobserved nodes				Unobserved intersections (number of nodes)				
		1	10	20	30	1 (4)	2 (8)	3 (12)	4 (16)	5 (20)
10 min	MAPE	3.89%	7.17%	7.80%	8.13%	6.25%	7.61%	8.34%	8.95%	9.47%
	RMSE	2.86	5.25	5.64	5.87	4.79	5.8	6.18	6.63	7.01
30 min	MAPE	5.11%	9.16%	9.62%	10.05%	7.46%	8.96%	9.82%	10.48%	11.06%
	RMSE	3.77	6.71	6.95	7.16	5.76	6.86	7.59	8.12	8.48
60 min	MAPE	9.81%	13.71%	14.37%	15.12%	11.15%	13.58%	14.67%	15.43%	16.08%
	RMSE	7.23	10.05	10.39	10.78	8.55	10.35	10.87	11.43	11.89

Figure 4. Evaluation Results

Evaluation

- Case 1: Speed prediction results when more than one direction is observed at the intersection.
- Case 2: Speed prediction results for the case where all intersection directions are not observed.
- Number of unobserved nodes and prediction time increases, the error rate increases significantly.

Conclusion & Future work

- We propose an urban traffic network modeling framework using CCTV data
- We plan to design a framework that can automatically reflect change in the recording area of CCTV cameras while optimizing the sampling frequency.

Extended DCRNN

- Add virtual nodes and edges in an incomplete traffic network with unobserved nodes.
- The DCRNN model trains the networks with T sequences, and predict the output networks for the $T+t$ sequence.
- An unobserved node is estimated through two adjacent virtual nodes.

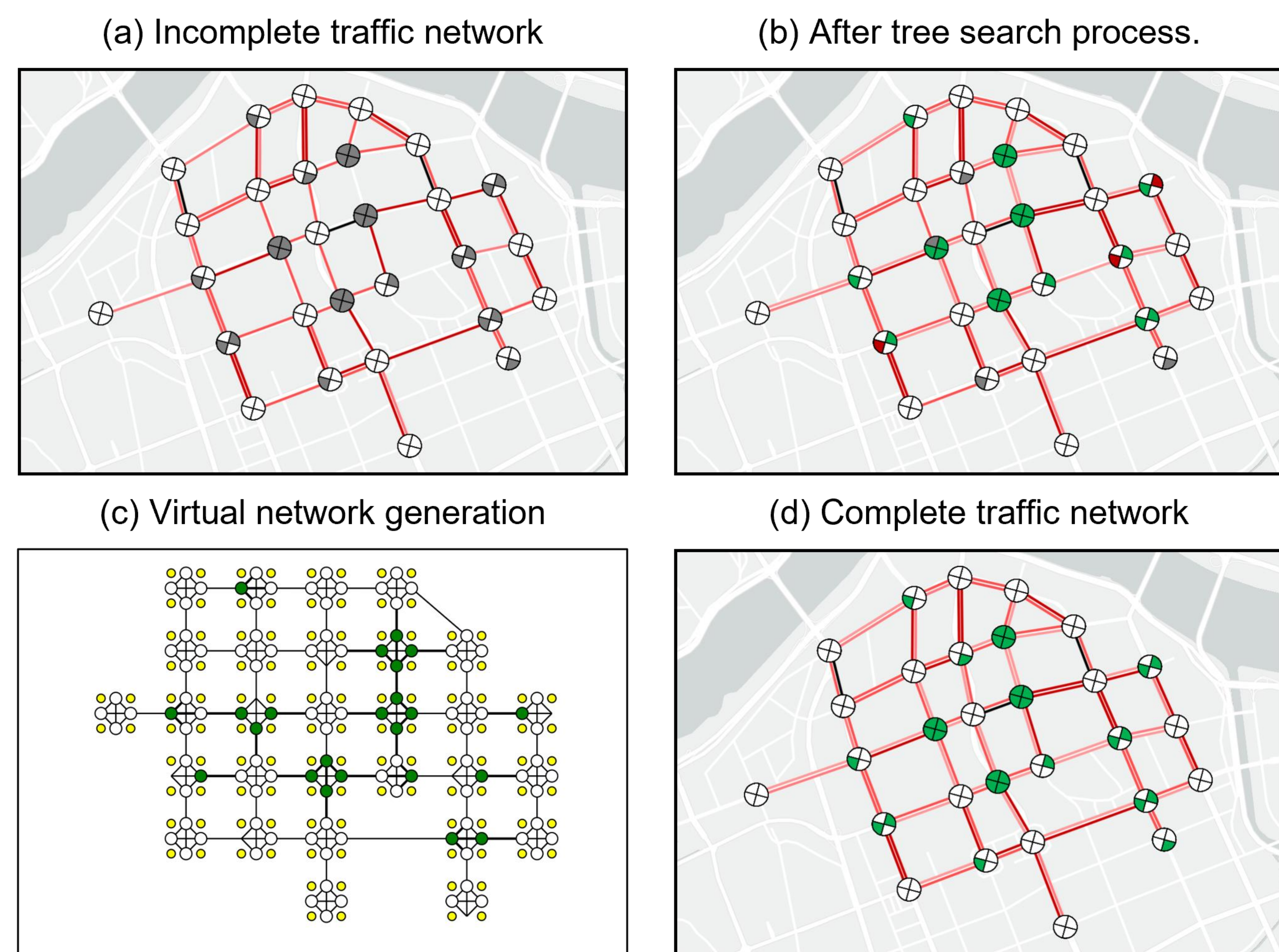


Figure 5. Case Study

Case study

- (a): An incomplete traffic network showing the vehicle speed in Seoul at 7 PM is extracted through the framework.
- (b): Tree search works interactively with a mouse click.
- (c): A virtual network for input data of extended DCRNN.
- (d): Estimate the unobserved nodes through the Extended DCRNN model.